Lab 5: Analog Signal Conditioning and Filter Design for Mechatronic Systems

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1- End-of-Session Shutdown Procedure

At the end of each lab session, please complete the following steps:

1. Return all resistors, op-amps, and other components to the plastic bag they came in, and check that nothing is missing before you leave.
2. Turn off the power supply.
3. Disconnect the DAQ from power and unplug the USB cable.
4. Shut down the laptop.
5. Place the laptop in the cabinet.

2- Submission Requirements and Deliverables

If you write a MATLAB code (mlx or slx) to answer a question, you need to upload the code file and its PDF with all the plots directly to D2L Dropbox.
All the figures must have appropriate labels. You can use MATLAB xlabel, ylabel, and legend commands.
To draw circuit diagrams, you may use online services like https://www.circuit-diagram.org
Handwritten diagrams are not accepted.

Everything in the table below is required. The answer type expected for each question is shown in square brackets after the question.

✓	What to submit	Where	When
	Individual exit survey	online	As posted on D2L
	Contribution table	In the report	As posted on D2L

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✓	What to submit	Where	When
	Report (PDF) with every figure, circuit diagram, photo and discussion asked for by the questions below	D2L Dropbox	As posted on D2L
	Simulink model files (.slx) for every model you build	D2L Dropbox	With the report
	MATLAB .mlx file(s) and their PDF export	D2L Dropbox	With the report
	Photographs of all circuits you design and build	In the report	With the report
	Circuit diagrams of all circuits you design and build (not freehand)	In the report	With the report

3- Materials

1. Provided laptop with MATLAB Signal Processing toolbox
2. QuanserQ8-USB data acquisition device (DAQ) and USB cable
3. Cabling
   Three custom RCA to 2 pins (signal and ground)
   The darker colour (Black/purple/blue) indicates the ground
4. Breadboard
5. Resistors: eight values. Identify what you actually have from the colour bands. Bags are not all identical, so work from the resistors in your own bag rather than assuming you have the same set as another group.
6. Capacitor (0.33 μF)
7. 741 Op-Amps (LM741CN op-amp, and the datasheet is uploaded to D2L)
8. power supply (the op-amp supply rails must be set before Section 5.4; the user manual is uploaded to D2L). Set channel 2 to +10 V at 1 A and channel 3 to -10 V at 1 A, which gives the op-amp supply rails of ±10 V. Do not raise either channel above 10 V.
9. jumper wires

4- Objective

This lab aims to provide a practical understanding of analog signal conditioning for mechatronic systems. You will design, build and test passive and active analog filters that separate a desired low-frequency signal from higher-frequency noise; evaluate filter performance in both the time domain and the frequency domain; compare each physical circuit against an equivalent s-domain model in Simulink; and investigate the capabilities and limitations of operational-amplifier circuits, including output saturation and the implementation of arithmetic operations on analog signals.

5- In-Lab Exercises

Setup procedure:
Required model configuration. Before you build any model in this lab, set the following. These settings are the same for every exercise unless a question says otherwise.

1. Connect the Q8-USB DAQ to the power bar with the power cable and to the laptop with the USB cable. Using the RCA-to-2-pin cables, connect Analog Output Channel #0 (AO #0) to the input of your circuit, and the output of your circuit to Analog Input Channel #0 (AI #0). The darker lead (black / purple / blue) is ground.
2. In the HIL Initialize block, set the board type to q8_usb (not qube_servo2_usb).
3. In Configuration Parameters → Solver, set the type to Fixed-step and the solver to ode1 (Euler).
4. Set the fixed-step size to 0.001 s, and set the Sample Time of the HIL Write and HIL Read blocks to 0.001 s.

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5. To get data out of Simulink and into MATLAB, use a To Workspace block with the save format set to Array, not Timeseries, and raise the data point limit (10000 is enough for this lab). Do the curve fits and all plotting in MATLAB, not in Simulink.
6. Sanity check: the highest frequency in this lab is 100 rad/s ≈ 15.9 Hz, so a 1 kHz sample rate is roughly 63 samples per noise period and about 31 times the Nyquist rate. Sampling is therefore not a limitation in this lab: anything you see in the data is the filter, not aliasing.

5.1 *Passive Analog Filter Design and Implementation*

Real-world signals are often contaminated with high-frequency noise. Filtering, either analog (using physical components) or digital (using mathematical algorithms), is essential for extracting clean signals from noisy data. In this lab, you will work with a signal containing a desired 10 rad/s component and unwanted 100 rad/s noise. You'll design and build passive RC filters using various resistor-capacitor combinations to determine optimal cutoff frequencies, then enhance your design using an operational amplifier to create an active filter. By comparing the frequency spectra of filtered outputs and evaluating performance, you'll gain practical insight into filter design principles, component selection criteria, and the advantages and limitations of passive versus active filtering approaches.

1. Set up a Simulink model that can produce three signals: $\sin(10t)$, $0.5\sin(100t)$, $\sin(10t) + 0.5\sin(100t)$. (1 point) [screenshot]
   Specifically, $\sin(10t) + 0.5\sin(100t)$ consists of both the signal you want to measure ($\sin(10t)$) and a higher frequency noise component with a lower Amplitude ($0.5\sin(100t)$).
   Your Simulink model should have an appropriate step size for the frequencies in the signal you are measuring.
   Looking at the formulas of these signals, fill out the following table. *[table]*

Signal	Period in seconds	Frequency in Hz	Circular frequency in rad/s	Amplitude in volts
$\sin(10t)$				
$0.5\sin(100t)$				

2. Plot the three signals. From the plots, measure the period in seconds, frequency in Hz, circular frequency in rad/s, and amplitude for each signal and report them here. Do they match what you expect? (2 points) *[figure + table]*

Signal	Period in seconds	Frequency in Hz	Circular frequency in rad/s	Amplitude in volts
$\sin(10t)$				
$0.5\sin(100t)$				

3. For $\sin(10t) + 0.5\sin(100t)$, plot the frequency spectrum. What are the frequencies in Hz and the corresponding amplitudes? Do they match the amplitudes and frequencies as expected? (2 points) *[figure + table]*

Signal	frequencies in Hz	Corresponding amplitudes in volts
$\sin(10t) + 0.5\sin(100t)$		

For the rest of this lab manual, only consider $\boldsymbol{\sin(10t) + 0.5\sin(100t)}$:

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4. To remove the noise from a signal, one can use a digital or analog filter. In this exercise, first, you need to build an analog filter using capacitors and resistors. If the higher frequency component of the signal is the noise, what is the type of filter that you need to build? Also, draw the circuit diagram for a passive filter of this type. (2 points) *[circuit diagram + 1-2 sentences]*

5. You have access to one capacitor and a number of different resistors. With these components, what are the cut-off frequencies for the filter that you could make? List the resistors and corresponding cutoff frequencies in the table. Which frequency do you think will work best for this situation? (3 points) *List the resistors that are actually in your own bag, identified from their colour bands — do not copy another group's values. [table + 2-3 sentences]*

Resistor in Ohm	Cutoff Frequency in rad/s	Cut off Frequency in Hz

6. Using the components and the breadboard, build the circuit you have designed. Put a photo of the circuit you have designed into your report. **Note:** if you are not confident in your circuit's design, ask a TA to check it. (1 point) *[photo]*
7. What is the transfer function of this filter?Give the transfer function in the form Vo/Vi = 1/(RCs+1) using your own R and C with values. *(1 point) [Derivation and final TF]*
8. Build a filter in the s-domain in Simulink with the same transfer function. *(1 point) [screenshot]*
9. Run your system, sending the signal through both the analog filter (physical circuit) and an equivalent s-domain model of the filter. Add a figure showing the signal and outputs of both filtered signals to your report. (2 points) *[figure]*
10. For the signals you receive from the analog and Simulink-based filters, generate a frequency spectrum and add it to your report. How do the spectrums compare with the original signal spectrum? (2 points) *[figure + 3-5 sentences]*
11. What would you need to do to convert your filter to a high-pass filter? Draw the circuit diagram of the new filter. Make the adjustments to your circuit and attach a photo of the new circuit. (3 points) *[circuit diagram + photo + 1-2 sentences]*
12. If you need to remove the low frequency component of the signal, what should be the value of the resistor using the same capacitor? (1 point) [number + 1-2 sentences]
13. If you are given a resistor that fits, use it; otherwise use the same resistor and run the filter in the high-pass configuration, then show the results. (1 point) *[figure]*
14. Find the transfer function of this filter, andbuild the Simulink block representing the transfer function of this filter. For the signals you receive from the analog and Simulink-based filters, generate a frequency spectrum and add it to your report. How do the spectra compare with the original signal spectrum? (2points) *[figure + 3-5 sentences]*

5.2 *Active Analog Filter Design and Implementation*

1. Given access to an op-amp as well as the original components, how could the filter be changed into an active filter from a passive filter? Draw the circuit diagram of the active low pass filter. (1 point) *[circuit diagram + 1-2 sentences]*
2. Set up the active filter circuit with your passive low-pass filter and run the same signal through it. Put both a picture of the circuit and the output signal into your report. Is there any change in amplitude? Is this what you expect from the magnitude plot of the filter? (2 points) *[photo + figure + 2-3 sentences]*
3. Generate the frequency spectrum or the active filter. How does this spectrum compare to the one from the passive filter? (2 points) *[figure + 2-3 sentences]*

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5.3 *Active vs. Passive Filters: Advantages and Limitations*

This section provides a direct comparative analysis of active and passive filter performance under conditions that highlight their distinct advantages and limitations. You need to modify signal conditions and filter configurations to experimentally observe key differences between passive and active implementations, including signal attenuation and amplification capabilities. Through systematic comparison of filter responses to weak signals dominated by noise, you will understand when each filter type is most appropriate for practical applications.

Procedure:

1. To observe the effects of an active filter, here, we use a much weaker signal: 0.1 sin(10t) with noise 0.5 sin(100t), where noise dominates. Now the noise (0.5) is five times larger than the signal (0.1).
2. Modify the active filter to have a gain of 5 or 10 (instead of unity gain). Explain how you can achieve this. This will amplify the filtered signal, demonstrating the amplification advantage. (1 point) *[circuit diagram + 2-3 sentences]*
3. Run the weak signal through both your passive low-pass filter and your active low-pass filter, and show both outputs together with the input on one figure. Plot the frequency spectrum of each and update the following table. (2 points) [figure + table]

	Frequencies in Hz	Corresponding amplitude in volts
Signal		
Filtered signal-active filter		
Filtered signal-passive filter		

4. What are the advantages of an active filter compared to a passive filter? Do you observe these here? (2 points) *[figure + 3-5 sentences]*

5.4 *Operational Amplifier Circuits and Limitations*

Operational amplifiers are fundamental building blocks in analog electronics, capable of performing mathematical operations such as addition, subtraction, multiplication by constants (gain), and integration on analog signals. In this section, you will explore both the capabilities and limitations of op-amps through hands-on experimentation. You will begin by constructing a basic non-inverting amplifier with unity gain to observe op-amp saturation limits, which are determined by the power supply voltages. Then, you will design a more complex op-amp circuit to implement a specific mathematical equation ($y = (2V_1 - V_2)/3$), requiring careful selection of resistor values and circuit topology. By comparing your circuit's actual output with theoretical predictions, you will identify practical factors that cause deviations from ideal behaviour.

Procedure:

1. Build a non-inverting op-amp circuit with a gain of 1. Put both a circuit diagram and a photo of the circuit into your lab report. **Note:** if you are not confident in your circuit's design, ask a TA or Lab Tech to check it. (2 points) *[circuit diagram + photo]*
2. Run a ramp signal sent from Simulink from -10 to 10 V through the circuit, and plot your output along with the input signal. What are the limits of the op-amp circuit, and what are these limits related to? (2 points) *[figure + 2-3 sentences]*

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3. Given the following mathematical equation, design a circuit to perform the calculation. For this circuit, you have access to up to 3 op-amps,and 10 resistors. Include a circuit diagram in the report and derive the final output given V1 and V2 as inputs. (2 points) *[derivation + circuit diagram]*

$$V_{out} = \frac{2V_1 - V_2}{3}$$

4. Build the op-amp circuit you have designed and put a photo of the complete circuit into your report. **Note 1:**The datasheet for the op-amps is posted on D2L and contains the pin assignments for setting up the circuit.**Note 2:** If you are not confident in your circuit's design, ask a TA to check it. (1 point) *[photo]*
5. Via Simulink, generate two time-varying signals V_1, V_2. Run the signal through your circuit to confirm that the output is what is expected. Report a plot of three signals: both the inputs and output signals. (2 points) *[figure]*
6. If there are any differences, what factors could be causing the difference between the signals from the theoretical result and your actual result? (1 point) *[2-3 sentences]*

6- References

1. Quanser Inc., "Quick Start Guide: Q8-USB Data Acquisition Device."
2. Texas Instruments, "LM741 Operational Amplifier" datasheet (posted on D2L).

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